

Hieu Tran

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RESEARCH INTERESTS

LLM Interpretability: representation engineering & activation editing (steering vectors) for reliable control

Agentic LLMs & Multi-Agent Systems: coding & web agents, coordination & tool use optimization

Machine Translation: multilingual transfer & synthetic data augmentation for low-resource MT

EDUCATION

University of Maryland, College Park

Maryland, USA

PhD Student in Computer Science, supervised by Marine Carpuat

Aug. 2024 – Present

- Achievements: Dean's Scholarship

VNUHCM - University of Science

Ho Chi Minh, Vietnam

Bachelor of Computer Science

Aug. 2017 – Jul. 2021

- Achievements: Outstanding Undergraduate Thesis Award

WORK EXPERIENCE

Senior Machine Learning Engineer

Jun. 2023 – Jul. 2024

BLACKBOX AI

Remote (Canada)

- Pioneered core features as an early engineer for flagship products, including the VSCode Coding Agent, Web Agent, Auto Completion, scaling the user base from their early stages over **30M users** and **4M VSCode** downloads
- Owned the end-to-end development of the code search/web search system, from researching ranking algorithms to implementing the final retrieval pipeline used in production
- Trained CodeLlama-34B on synthetic instruction-following datasets, achieving **86.6%** on HumanEval, surpassing GPT-4 (Nov 2023) which scored 85.4%

Lecturer & Reseach Collaborator

May. 2021 – Aug. 2024

VietAI

Ho Chi Minh, Vietnam

- Taught Foundations of Deep Learning, Advanced NLP, and LLMs courses to over **200 students**
- Worked directly with Dr. Minh-Thang Luong and Dr. [Trieu H. Trinh](#) (Google DeepMind) to build English-Vietnamese machine translation systems that outperform previous SOTA by 4.5% (BLEU).
- Released the first Vietnamese generative models, achieving SOTA in Summarization and Biomedical benchmarks.

AI Research Engineer

Aug. 2021 – Aug. 2023

M_Service (MoMo) (3rd unicorn in Vietnam)

Ho Chi Minh, Vietnam

- Built Intent-based/ChatGPT-based AI chatbots for Customer Support on MoMo #1 e-wallet platform in Vietnam
- Managed the training pipeline and chatbots on Kubernetes deployed on Google Cloud Platform
- Trained base language models that outperformed open-source Vietnamese models for our internal needs

Undergraduate Research Student

Sep. 2020 – Feb. 2022

AISIA Research Lab

Ho Chi Minh, Vietnam

- Explored the efficiency of Transformer-based models for Vietnamese Punctuation Prediction

AI Software Engineering Intern

Apr. 2019 – Aug. 2021

Zalo Group - VNG Corporation (1st unicorn in Vietnam)

Ho Chi Minh, Vietnam

- Constructed RDF-triplestore Knowledge Bases for Kiki - the first Vietnamese AI virtual assistant
- Proposed a light and state-of-the-art Vietnamese Spelling Corrector. [Try it!](#)

TECHNICAL SKILLS

Languages: Python, NodeJS, Typescript, SQL

Developer Tools: Git, Docker, Google Cloud Platform, MongoDB, PostgreSQL, Kubernetes, ElasticSearch

Libraries: Pytorch, Tensorflow, Transformers, DeepSpeed, PEFT, scikit-learn, Pandas, Numpy, Flask

PUBLICATIONS

1. **Hieu Tran***, Phuong-Anh Nguyen-Le*, Huy Nghiem, Quang-Nhan Nguyen, Wei Ai, Marine Carpuat. 2025. VietMix: A Naturally Occurring Vietnamese-English Code-Mixed Corpus with Iterative Augmentation for Machine Translation. *Submitted to EACL 2025*
2. Long Phan*, Tai Dang*, **Hieu Tran***, Trieu H. Trinh*, Vy Phan, Lam D. Chau, Minh-Thang Luong. 2023. Enriching Biomedical Knowledge for Low-resource Language Through Large-Scale Translation. In *EACL 2023*
3. Chinh Ngo*, Trieu H. Trinh*, Long Phan*, **Hieu Tran***, Hieu Nguyen, Tai Dang, Minh Nguyen, Minh-Thang Luong. 2022. MTet: Multi-domain Translation for English and Vietnamese. In *ArXiv Preprint*
4. Long Phan, **Hieu Tran**, Hieu Nguyen and Trieu H. Trinh. 2022. ViT5: Pretrained Text-to-Text Transformer for Vietnamese Language Generation. In *Proceedings of NAACL SRW 2022*
5. **Hieu Tran**, Long Phan, James T. Anibal, Binh T. Nguyen and Truong-Son Nguyen. 2021. SPBERT: An Efficient Pre-training BERT on SPARQL Queries for Question Answering over Knowledge Graphs. In *Proceedings of ICONIP 2021*, Acceptance rate $\sim 20\%$
6. Long Phan, **Hieu Tran**, Daniel Le, Hieu Nguyen, James T. Anibal, Alec Peltekian and Yanfang Ye. 2021. CoTexT: Multi-task Learning with Code-Text Transformer. In *Proceedings of NLP4Prog @ ACL-IJCNLP 2021*
7. **Hieu Tran**, Cuong Viet Dinh, Long Phan and Son Nguyen. 2021. Hierarchical Transformer Encoders for Vietnamese Spelling Correction. In *Proceedings of IEA/AIE 2021*
8. **Hieu Tran**, Cuong Viet Dinh, Quang Hong Pham and Binh T. Nguyen. 2021. An Efficient Transformer-Based Model for Vietnamese Punctuation Prediction. In *Proceedings of IEA/AIE 2021*
9. Long Phan, James T. Anibal, **Hieu Tran**, Shaurya Chanana, Erol Bahadroglu, Alec Peltekian and Grégoire Altan-Bonnet. 2021. SciFive: a text-to-text transformer model for biomedical literature. *ArXiv Preprint*

PROJECTS

Coding Agent on Browser | *Typescript, APIs, CodeSandbox*

- Built a browser UI for user-submitted coding tasks; an LLM agent follows the *ReAct* framework to plan steps and decide which tools to call
- Executes a chain of tools to operate on the project file system—edit/read/search files and run a sandboxed terminal—to iteratively complete tasks end-to-end

Multi-Agent Machine Translation System | *Python, Transformers, LiteLLM*

- Designed a multi-agent framework that orchestrates specialized LLM agents (*ResearchAgent*, *DraftAgent*, *RefineAgent*, *EvaluationAgent*) to collaboratively perform translation for low-resource languages

Activation Steering for Style Transfer | *Python, Pytorch, Transformers*

- Extracted style activation vectors and applied activation editing at inference to steer LLM outputs toward user-specific styles while preserving semantics

TALKS AND ACTIVITIES

The Art of Generative AI, talk at Google Developer Student Clubs - IU 2024, 100+ audiences

Unlocking the potential of GenAI for coding, talk at Google Devfest HCMC 2023, 100+ audiences

Explore the Power of AI - Career Trends in This Industry, USAID WISE Online Event 2023, 500+ audiences

Shaping Your NLP Career Path in 2024, talk at VietAI Online Event 2023, 150+ audiences

Challenges and Opportunities When Working In AI Industry (2022), Van Lang University, 100+ audiences

MTet: Multi-domain Translation for English and Vietnamese, talk at Google I/O Extended Hanoi 2022

How MoMo Builds an AI Chatbot, published on MoMo Technology Blog (Medium)

Participated in **BigScience Research Workshop**, mainly organized by HuggingFace